Shril Kumar

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Senior Software Engineer

Software engineer with 8+ years building production-scale Data Platforms and AI infrastructure. Experienced in high-volume data pipelines, low-latency services, agentic data access, data governance, quality, lineage and observability across data estates up to 70 PB processing 1B+ events daily.

Experience

- **Roku** — Bangalore, IN
  Senior Software Development Engineer (Data) — *September 2024 – Present*
  - **Ads Data Platform Architecture**: Led a team-wide overhaul of impression, conversion, and attribution ingestion service in Scala Spark for **200M ad impressions and 50M advertiser conversion events daily** under a one-hour publishing SLA; replaced external tracker state and non-idempotent Hive pipelines with stateless, idempotent Iceberg orchestration using partition-held state, Bloom filters, and positional deletes.
    - **Quality, Observability, and Lifecycle**: Cut query execution runtime by **35% across 12 hourly Spark ML Apps** and **20%** for Ad-Hoc Trino queries; introduced transactional write-audit-publish controls and safe reruns, reducing incident recovery from **1 day to 2 hours**.
  - **Data-Query MCP**: Built and productionised a governed MCP interface so agents and engineers can query enterprise data through DataHub and Trino only with certified metrics, schema, lineage, and ownership context. Implemented per-user OAuth scoping, JWT validation, ownership-aware access errors, certified-table ranking, and caller-role audit logs; adopted across Roku's **120+ engineer** Data Platform organisation, handling approximately **5,000 daily calls** over approximately **5,000+ tables**.
  - **Iceberg Table Maintenance Framework**: Designed a config-driven framework in Rust to automate Iceberg maintenance, including data-file and positional-delete compaction, snapshot management, and orphan cleanup, delivering compaction comparable to Apache Spark at **90% lower compute cost**; now maintains **200+ production tables**.
  - **PII Discovery and Evaluation-to-Data Loop**: Proposed and shipped an in-house PII discovery engine instead of a planned **$150K/year** third-party tool. Clustered manual overrides and validation failures into **6 failure modes**, generated provenance-tagged synthetic examples for the weakest clusters, and served them through retrieval-augmented few-shot prompting raising precision from **84% to 91% at recall** $\geq 99\%$ on a 3,000-column governance-validated dataset.
  - **Frost Governed Migration**: Built and productionised an autonomous Hive-to-Iceberg migration system that discovers lineage, ownership, producing workflows, write patterns, and PII/SOX classifications; risk-tiered datasets and enforced three human-review gates with joint Data Engineering and Data Platform approval. Migrated **150 tables across 40 workflows**, with **75% of non-critical tables requiring no engineering tuning** after mandatory review and **no production incidents**.
  - **GDPR Data Deletion**: Led the redesign of deletion workflows across a **70 PB** data lake, introducing anonymisation and configurable Spark processing; reduced end-to-end fulfilment from **90+ days to 18 days (80%)** and separately purged deeply nested PII from **700+ TB** of Hive data.
  - **GDPR Data Download**: Re-architected Roku's GDPR data-download workflow from MapReduce to Spark, raising request-intake capacity from **20 to 1,000+/day** and using customer/time partition pruning against a **20 PB** one-year source footprint, with daily physical reads estimated at approximately **200 TB**; worked with legal analysts to make new compliance data types onboardable through configuration instead of engineering work.
  - **Pager Incident Response Agent**: Designed and productionised Slack incident-history and Confluence runbook-retrieval MCP servers for an internal incident-management assistant, with read allowlists, least-privilege scopes, button-gated writes, and Bazel, Terraform, and Kubernetes deployment; adopted by **20+ engineering teams and 150+ engineers**.
- **Groupon** — Bangalore, IN
  Software Development Engineer - III — *Oct 2020 – August 2024*
  Tech Lead — *September 2023 – August 2024*
  - **Personalization Service**: Designed and built a Java/Dropwizard batch and real-time personalization service supporting **40M emails and 60M push notifications daily**, with **5 ms p95** end-to-end latency.
  - **Consumer 360 Data Store**: Built a **200 TB+** Consumer 360 data store with **400+** behavioral, demographic, and predictive features used by Marketing Engineering.

- **Dataproc Migration**: Led the architecture and phased cutover of **2,500 Spark jobs** from on-premises infrastructure to Google Cloud Dataproc; designed bidirectional table synchronisation to preserve end-to-end marketing flows during per-job cutovers without interrupting revenue paths.
- **Keyspaces-to-Bigtable Migration**: Led a no-downtime migration from Amazon Keyspaces to Cloud Bigtable, cutting annual run cost **90%**, from approximately **$2M to $200K**.
- **Reusable Datastore Libraries**: Architected bulk-CRUD libraries for Amazon Keyspaces and Cloud Bigtable using batching and parallel execution; adopted as the standard access path by **10+ teams**.
- **Data Quality Framework**: Designed record-, column-, and metric-level checks with alerting so warehouse pipelines surface correctness and freshness failures early; reduced data-corruption incidents tracked in Jira by **75% over two quarters**.
- **Data Lifecycle Management**: Implemented retention and lifecycle policies in the data warehouse, reducing retained data volume by **80%** and annual storage cost by approximately **$500K**.

- **AtkinsRéalis** Bangalore, IN
  Software Development Engineer - II *Feb 2019 – Sep 2020*
  - Built image and GIS data pipelines on a Databricks-based big-data and machine-learning platform, and optimised PySpark workloads using Spark SQL and UDFs.
  - Built serverless ETL applications that moved MongoDB data into Azure SQL Server for the Assist 2.0 workflow product.

- **Eatclub** Bangalore, IN
  Software Development Engineer - I *Jun 2018 – Jan 2019*
  - Built backend APIs for vendor onboarding and inventory tracking, including QR-code integrations with Box8's e-commerce platform.

Skills

- **Languages**: Python, SQL, Scala, Java
- **Data Infrastructure**: Apache Spark, Iceberg, Trino, Hive, Kafka, Hadoop, Temporal, Airflow, Databricks
- **Governance & Data Quality**: PII classification, provenance, lineage, access controls, audit logging, retention/deletion, DataHub
- **AI Data Systems**: synthetic data, failure clustering, retrieval-augmented prompting, LLM-based classification, FastMCP / MCP
- **Cloud, Infrastructure & Datastores**: GCP (Dataproc, Bigtable), AWS (Keyspaces), Terraform, Kubernetes, Bazel, Cassandra, PostgreSQL, DynamoDB

Open Source Contributions

- **Hardwood**: Contributed to Hardwood, a minimal-dependency Java implementation of Apache Parquet: authored dictionary-space predicate evaluation; restored Bloom filter pruning on `FLOAT`/`DOUBLE` columns; and packed the `dive` TUI preview by content width.
- **Koalas (Databricks)**: Contributed merged DataFrame APIs (replace, cache, to_records, to_excel) to Koalas, later merged into **Apache Spark** as the pandas API on Spark.

Education

- **Indian Institute of Technology (Indian School of Mines), Dhanbad** Dhanbad, India
  B.Tech in Engineering; Minor in Computer Science and Engineering; CGPA: 8.1/10 *Jul 2014 – Mar 2018*

Internships

- SNC-Lavalin Nuclear Solutions (2017) • AtkinsGlobal Digital Incubator (2016) • Saint Petersburg State University, Global Citizen Fellow (2015)